The conversion and process conversion measures necessary to comply with this revision shall be completed by 7 October 1988.

MIL-S-19500/565A
7 October 1987
SUPERSEDING
MIL-S-19500/565
24 January 1986

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, P-CHANNEL, SILICON, TYPES 2N6895, 2N6896, 2N6897 AND 2N6898 JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a P-Channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1 and 2; TO-205AF for type 2N6895; TO-204AA for 2N6896 and 2N6897; and TO-204AE for 2N6898.

1.3 Maximum ratings.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} T_C = +100°C	I_S 2/	I_{DM}	T_{op} and T_{stg}
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C
			100	100	±20					
2N6895	8.33	.6				1.16	0.74	1.16	5	
2N6896	60	4				6.0	3.8	6.0	20	-55
2N6897	100	4				12	7.6	12	30	to
2N6898	150	4				25	15.8	25	60	+150

1/ Derate linearity T_C > +25°C - 2N6895 (0.067 W/°C), 2N6896 (0.48 W/°C), 2N6897 (0.8 W/°C), 2N6898 (1.2 W/°C).
2/ Derate above T_C = +25°C according to the formula $I_D = \sqrt{\frac{P(rated)}{k}}$, where $P(rated) = P_T - (T-25)$ (W/°C) watts;
K = max $r_{DS(on)}$ at T_J = +150°C.

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = -1 mA	$V_{GS(th)}$ 1 $V_{DS} \geq V_{GS}$ I_D = -1 mA	Max I_{DSS1} V_{GS} = 0 V	$R_{\theta JC}$	Max $r_{DS(on)}$ 1/ V_{GS} = -10 V dc	
			V_{DS} = 80% of rated V_{DS}		T_J = +25°C at I_{D1}	T_J = +150°C at I_{D2}
	V dc	V dc	μA dc	°C/W	Ohms	Ohms
	-100	Min -2.0 Max -4.0	-1.0			
2N6895				15.0	3.65	6.15
2N6896				2.083	0.6	1.67
2N6897				1.25	0.3	0.69
2N6898				0.83	0.2	0.24

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302, by using the self-addressed Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

$V_{(BR)DSS}$ - Drain to source breakdown voltage, all other terminals short-circuited to source.

I_S - Source current through drain diode (forward biased V_{SD}).

C - Coulomb.

g_{FS} - DC forward transconductance.

$I_{(ISO)}$ - Source pin to case isolation current.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

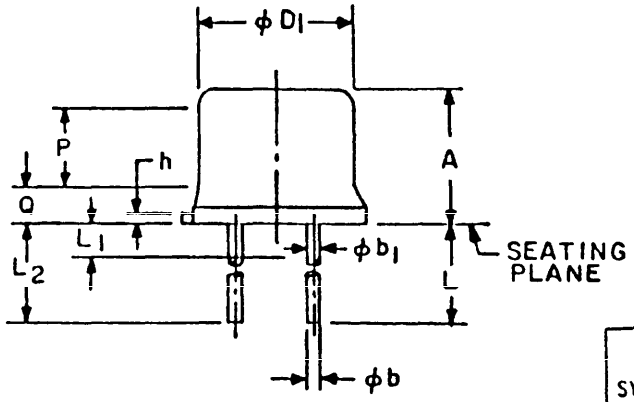
3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. A copper core is permitted. Lead finish shall be gold, tin, or solder. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.3).

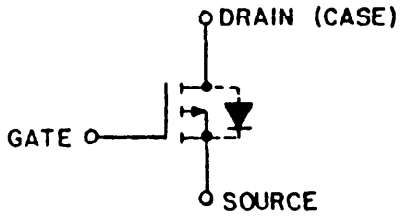
3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

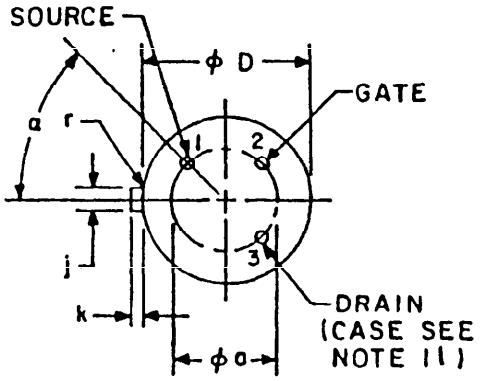
3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection (see 6.2).

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Circuit schematic.



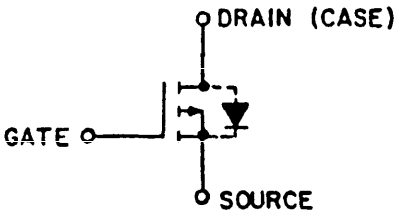
SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	Min	Max	Min	Max	
A	.160	.180	4.07	4.57	
ϕa	.200 TP		5.08 TP		6
ϕb	.016	.021	0.41	0.53	7,8
ϕb_1	.016	.019	0.41	0.48	7,8
ϕD	.335	.370	8.51	9.40	
ϕD_1	.305	.335	7.75	8.51	
h	.009	.041	0.23	1.04	
j	.028	.034	0.71	0.86	2
k	.029	.045	0.74	1.14	3
L	.500	.750	12.70	19.05	7,8
L_1		.050		1.27	7,8
L_2	.250		6.35		7,8
P	.100		2.54		5
Q		.050		1.27	4
r		.010		0.25	9
a	45 TP		45 TP		6

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond radius r maximum, j shall be held for a minimum length of .011 (0.028 mm).
4. k measured from maximum ϕD.
5. Outline in this zone is not controlled.
6. ϕD_1 shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown on figure 3.
8. ϕb_1 applies between L_1 and L_2. ϕb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
9. All three leads.
10. Radius r applies to both inside corners of tab.
11. Drain is electrically connected to case.

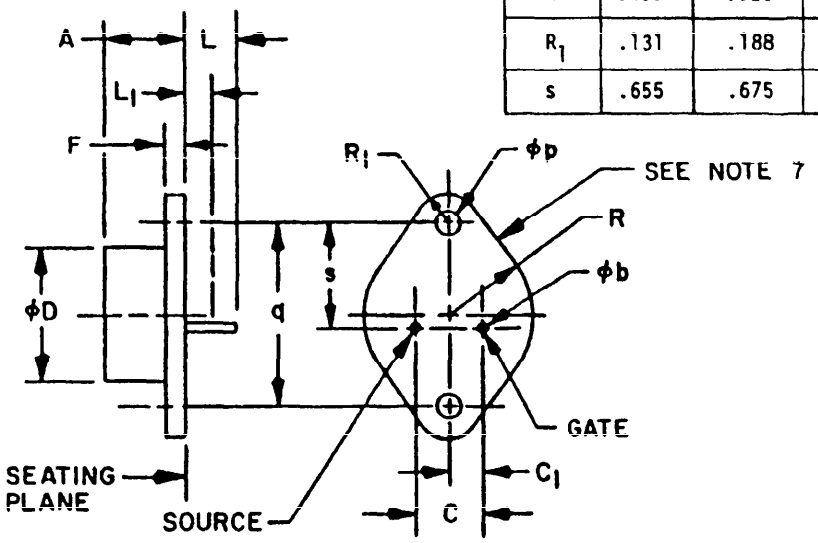
FIGURE 1. Physical dimensions of type 2N6895.

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Circuit schematic.

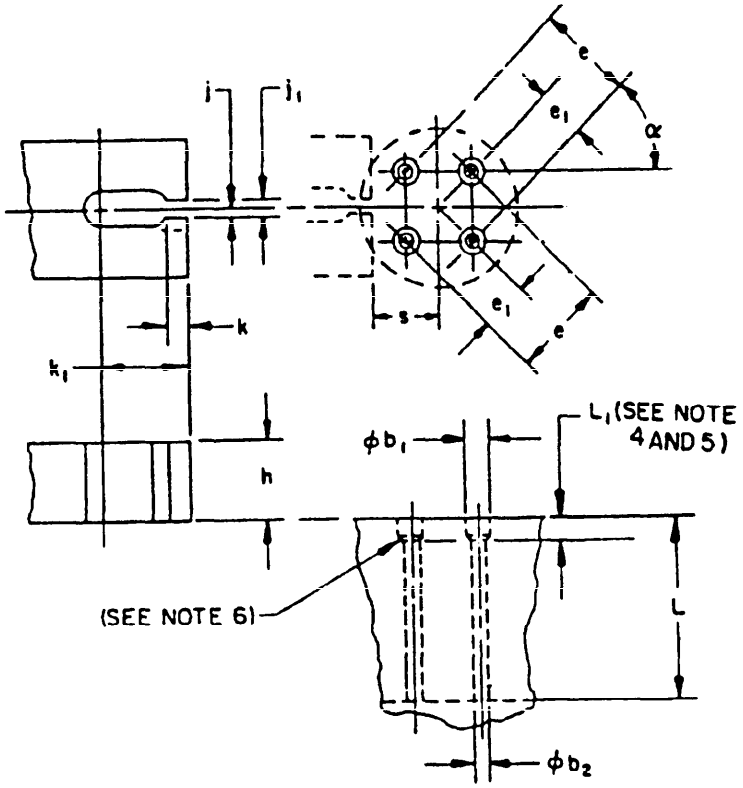
SYMBOL	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	Min	Max	Min	Max	
A	.250	.360	6.35	9.15	
ϕb	.057	.063	1.45	1.60	5
ϕb	.038	.043	0.97	1.10	6
ϕD		.875		22.23	
C	.420	.440	10.67	11.18	1
C_1	.205	.225	5.21	5.72	1
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1		.050		1.27	
ϕp	.151	.161	3.84	4.09	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	



NOTES:
1. These dimensions should be measured at points .050 (1.27 mm) .055 (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
2. The seating plane of the header shall be flat within .001 (0.03 mm) concave to .004 (0.10 mm) convex inside a .930 (23.62 mm) diameter circle on the center of the header and flat within .001 (0.03 mm) concave to .006 (0.15 mm) convex overall.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. These dimensions pertain to the 2N6898 type.
6. These dimensions pertain to the 2N6896 and 2N6897 types.
7. The drain is electrically connected to the case.

FIGURE 2. Physical dimensions of transistor type 2N6896, 2N6897, 2N6898, TO-204AA and TO-204AE.

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Ltr	Dimensions				
	Inches		Millimeters		
	Min	Max	Min	Max	Notes
ϕb_1	.0595	.0605	1.51	1.54	
ϕb_2	.0325	.0335	0.83	0.85	
e	.1995	.2005	5.07	5.09	
e_1	.0995	.1005	2.53	2.55	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	0.44	0.46	
j_1	.0350	.0355	0.89	0.90	
k	.009	.011	0.23	0.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	44.10°	44.90°	45.15°	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
4. Gauging procedure. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±.25 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge.
5. Gauging plane.
6. Drill angle.

FIGURE 3. Gauge for lead and tab locations.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	Test condition F, except T_{low} = -55°C, 20 cycles	Test condition F, except T_{low} = -55°C, 20 cycles
1/ 2/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9	I_{GSS1}, I_{DSS1}, gate stress test (see 4.5.5) Subgroup 2 of table I herein	Gate stress test (see 4.5.5) Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	Subgroup 2 of table I herein I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±.2 µA dc or ±100% of initial value, whichever is greater	Subgroup 2 of table I herein I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$
12	Method 1042, Test condition A and test condition C (see 4.3.1)	Method 1042, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±.2 µA dc or ±100% of initial value, whichever is greater $\Delta r_{DS(on)1}$ = ±20% of initial value $\Delta V_{GS(th)1}$ = ±20% of initial value	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±.2 µA dc or ±100% of initial value, whichever is greater $\Delta r_{DS(on)1}$ = ±20% of initial value $\Delta V_{GS(th)1}$ = ±20% of initial value

1/ Shall be performed anytime before screen 9.
2/ This test method in no way implies an avalanche energy rating. This is a stress test designed to ensure a rugged product.

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4.3.1 Power burn-in. Power burn-in conditions are as follows: T_A = +25°C, -5°C +10°C; $V_{DS} \geq 10$ V minimum. I_D is adjusted to meet a junction temperature of 140°C, -0°C, +10°C. By controlling V_{GS} voltage, one can obtain the specific I_D current required to apply the appropriate power for the device under stress. Power condition and T_J requirements can be established by the ΔV_{SD} measurement technique described in MIL-STD-750, method 3161. For condition C, t = 240 hours.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end po'nts) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal impedance. Thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC}$(max) = 15.0°C/W (2N6895), 2.083°C/W (2N6896), 1.25°C/W (2N6897), and 0.83°C/W (2N6898).

2N6895		2N6896	
I_M measuring current	10 mA	I_M measuring current	10 mA
I_H drain heating current	.6 A	I_H drain heating current	2 A
t_H heating time	steady state 1/	t_H heating time	steady state 1/
V_H drain-source heating voltage	10 V	V_H drain-source heating voltage	20 V
t_{MD} measurement time delay	10 to 80 µs	t_{MD} measurement time delay	10 to 80 µs
t_{SW} sample window time	10 µs maximum	t_{SW} sample window time	10 µs maximum

2N6897		2N6898	
I_M measuring current	10 mA	I_M measuring current	10 mA
I_H drain heating current	3.5 A	I_H drain heating current	4 A
t_H heating time	steady state 1/	t_H heating time	steady state 1/
V_H drain-source heating voltage	20 V	V_H drain-source heating voltage	25 V
t_{MD} measurement time delay	10 to 80 µs	t_{MD} measurement time delay	10 to 80 µs
t_{SW} sample window time	10 µs maximum	t_{SW} sample window time	10 µs maximum

4.5.3 Thermal response (ΔV_{SD} measurements). The delta V_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 4) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

2N6895		2N6896	
I_M measuring current	10 mA	I_M measuring current	10 mA
I_H drain heating current	.6 A minimum	I_H drain heating current	2 A minimum
t_H heating time	10 ms	t_H heating time	100 ms
V_H drain-source heating voltage	10 V minimum	V_H drain-source heating voltage	20 V minimum
t_{MD} measurement time delay	10 to 80 µs	t_{MD} measurement time delay	10 to 80 µs
t_{SW} sample window time	10 µs maximum	t_{SW} sample window time	10 µs maximum

1/ (See MIL-STD-750 method 3161 for definition.)

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2N6897		2N6898	
I_M measuring current - - - - - -	10 mA	I_M measuring current - - - - - -	10 mA
I_H drain heating current - - - -	3.5 A minimum	I_H drain heating current - - - -	4 A minimum
t_H heating time - - - - - - - - -	100 ms	t_H heating time- - - - - - - - -	100 ms
V_H drain-source heating voltage-	20 V minimum	V_H drain-source heating voltage-	25 V minimum
t_{MD} measurement time delay - - -	10 to 80 µs	t_{MD} measurement time delay - - -	10 to 80 µs
t_{SW} sample window time - - - - -	10 µs maximum	t_{SW} sample window time - - - - -	10 µs maximum

4.5.4 Unclamed inductive switching.

Peak current (I_D):

2N6895 - - - - - - - - - - - 1 A
2N6896 - - - - - - - - - - - 2.7 A
2N6897 - - - - - - - - - - - 5 A
2N6898 - - - - - - - - - - - 5.9 A

Peak gate voltage (V_{GS}) - - - - - - - - - 10 V

Gate to source resistor (R_{GS})

Initial case temperature (T_C) - - - - - 25°C +10%, -5%

Inductance (L)- - - - - - - - - - - - - - 100 µH ±10%

Number of pulses to be applied- - - - - 1

4.5.5 Gate stress test.

V_{GS} = 30 V minimum

t = 250 µs minimum

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 1								
Visual and mechanical inspection	2071							
Subgroup 2								
Breakdown voltage, drain to source (threshold)	3407	I_D = -1.0 mA dc bias condition C, V_{GS} = 0 V;			$V_{(BR)DSS}$	-100		V dc
Gate to source voltage	3403	$V_{DS} \geq V_{GS}$, I_D = -0.25 mA dc			$V_{GS(th)1}$	-2.0	-4.0	V dc
Gate current	3411	V_{GS} = ±20 V dc, bias condition C, V_{DS} = 0 V			I_{GSS1}		±100	nA
Drain current	3413	V_{DS} = -80 V dc, bias condition C, V_{GS} = 0 V			I_{DSS1}		-1.0	µA
Static drain to source "on" - state resistance	3421	V_{GS} = -10 V dc, pulsed (see 4.5.1), condition A			$r_{DS(on)1}$			
2N6895		I_D = -0.74 A dc					3.65	Ohm
2N6896		I_D = -3.8 A dc					0.6	Ohm
2N6897		I_D = -7.6 A dc					0.3	Ohm
2N6898		I_D = -15.8 A dc					0.2	Ohm
Drain to source "on" - state voltage	3405	V_{GS} = 10 V dc, pulsed (see 4.5.1), condition A			$V_{DS(on)1}$			
2N6895		I_D = -1.16 A dc					-6.0	V
2N6896		I_D = -6.0 A dc					-6.0	V
2N6897		I_D = -12.0 A dc					-4.8	V
2N6898		I_D = -25.0 A dc					-6.0	V
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V			V_{SD}			
2N6895		I_S = -1.16 A dc				-0.8	-1.6	V
2N6896		I_S = -6.0 A dc				-0.8	-1.6	V
2N6897		I_S = -12.0 A dc				-0.8	-1.6	V
2N6898		I_S = -25.0 A dc				-0.8	-1.6	V
Forward transconductance	3475	Pulsed (see 4.5.1)			g_{FS}			
2N6895						0.2		S
2N6896						1.0		S
2N6897						2.0		S
2N6898						4.0		S

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 3								
High temperature operation:		$T_C = T_J = +125°C$						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = -0.25$ mA dc			$V_{GS(th)2}$	-1.0		V dc
Gate reverse current	3411	$V_{GS} = \pm 20$ V dc, bias condition C, $V_{DS} = 0$ V			I_{GSS2}		±200	nA dc
Drain current	3413	$V_{DS} = -80$ V dc, bias condition C; $V_{GS} = 0$			I_{DSS2}		-50	µA dc
Static drain to source on-state resistance	3421	$V_{GS} = -10$ V dc, pulsed (see 4.5.1)			$r_{DS(on)2}$			
2N6895		$I_D = -0.74$ A dc					5.66	Ohm
2N6896		$I_D = -3.8$ A dc					0.96	Ohm
2N6897		$I_D = -7.6$ A dc					0.465	Ohm
2N6898		$I_D = -15.8$ A dc					0.22	Ohm
Low temperature operation:		$T_C = T_J = -55°C$						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; $I_D = -.25$ mA dc			$V_{GS(th)3}$		-5.0	V dc
Subgroup 4								
Switching time test	3472	I_D = half rated I_{D1}, $V_{GS} = 10$ V dc, $R_{gen} = 15\ \Omega$, $R_{GS} = 15\ \Omega$, V_{DD} = 50% of rated V_{DS}						
Turn-on delay time		$V_{DD} = -50$ V dc			$t_{d(on)}$			
2N6895		$I_D = -0.74$ A dc				---	25	ns
2N6896		$I_D = -3.8$ A dc				---	60	ns
2N6897		$I_D = -7.6$ A dc				---	60	ns
2N6898		$I_D = -15.8$ A dc				---	50	ns
Rise time		$V_{DD} = -50$ V dc			t_r			
2N6895		$I_D = -0.74$ A dc				---	45	ns
2N6896		$I_D = -3.8$ A dc				---	100	ns
2N6897		$I_D = -7.6$ A dc				---	175	ns
2N6898		$I_D = -15.8$ A dc				---	250	ns
Turn-off delay time		$V_{DD} = -50$ V dc			$t_{d(off)}$			
2N6895		$I_D = -0.74$ A dc				---	45	ns
2N6896		$I_D = -3.8$ A dc				---	150	ns
2N6897		$I_D = -7.6$ A dc				---	275	ns
2N6898		$I_D = -15.8$ A dc				---	400	ns

See footnote at end of table.

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or air propagated outward from a nuclear explosion in which th

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 4 - continued								
Fall time		V_{DD} = -50 V dc			t_f			
2N6895		I_D = -0.74 A dc				---	50	ns
2N6896		I_D = -3.8 A dc				---	100	ns
2N6897		I_D = -7.6 A dc				---	175	ns
2N6898		I_D = -15.8 A dc				---	250	ns
Subgroup 5								
Safe operating area	3474	See figure 4						
High voltage test		V_{DS} = 80% of rated V_{DS}						
2N6895								
2N6896								
2N6897								
2N6898								
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7						
Subgroup 6								
Not applicable								
Subgroup 7								
Gate charge	3471	Condition A and B						
Test 1								
Minimum off-state gate charge					$Q_{g(th)}$			nC
2N6895						.17	.33	
2N6896						.73	1.4	
2N6897						1.5	2.9	
2N6898						3.0	5.7	
Test 2								
On-state gate charge					$Q_{g(on)}$			nC
2N6895						2.5	4.7	
2N6896						13	24	
2N6897						31	56	
2N6898						63	117	

See footnote at end of table.

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b. "commercial-type" items (commercially designed and built items modified to meet

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 7 - continued								
Test 3								
Maximum on-state gate charge					$Q_{gm(on)}$			nC
2N6895						4.7	9.2	
2N6896						25	46	
2N6897						56	104	
2N6898						114	212	
Test 4								
Gate plateau voltage					V_{GP}			V
2N6895						4.9	9.5	
2N6896						4.9	9.2	
2N6897						4.2	8.0	
2N6898						4.1	7.8	
Test 5								
Gate to source charge					Q_{gs}	.4	1.2	nC
						1.1	5.5	
2N6895						3	13	
2N6896						6	25	
2N6897								
2N6898								
Test 6								
Gate to drain charge					Q_{gd}			nC
2N6895						.9	2.9	
2N6896						5.5	14.5	
2N6897						14	36	
2N6898						26	69	
Reverse recovery time	3473	$V_{DD} \leq 30$ V $di/dt = 100$ A/µs			t_{rr}			
2N6895		$I_F = 4$ A					340	ns
2N6896		$I_F = 4$ A					375	ns
2N6897		$I_F = 4$ A					500	ns
2N6898		$I_F = 4$ A					750	ns

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figures 1 and 2		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051	Test condition F3, except T_{low} = -55°C		
Hermetic seal Fine leak Gross leak	1071			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		
Decap internal design verification	2075	See 3.3.2		
SEM	2077			
Bond strength	2037	Test condition A; all internal wires for each device shall be pulled separately		
Die shear	2017			
Subgroup 4				
Intermittent operation life	1042	Test condition D, see 4.3.1. Power on time shall be 3 minutes minimum 2,000 cycles. No heat sink or forced-air cooling on the device shall be permitted, during the on cycle		
2N6895		V_{DS} = -10 V dc; P_T = .6 W at T_A = 25°C ±3°C;		
2N6897, 2N6898		V_{DS} = -20 V dc; P_T = 4 W at T_A = 25°C ±3°C		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8		

See footnote at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection LTPD 1/	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5				
Accelerated steady-state operation life	1042	120 hours; See 4.3.1 T_J = 200°C Marking legibility requirements shall not apply.		
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7		
Bond strength (Al-Au die interconnects only)	2037	Test condition A		
Subgroup 6				
Thermal impedance	3161	See 4.5.2		

1/ For LTPD numbers, see MIL-S-19500.

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This design must satisfy all requirements of

TABLE IIb. Group B inspection for JANTX, JANTXV.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051	Test condition F1, except T low = -55°C, 25 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			
Intermittent operation life (LTPD)	1042	Test condition D, 2,000 cycles A cycle shall be 30 second minimum for the 2N6895 and 1 minute minimum for the 2N6896, 2N6897, and 2N6898.	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8	
Subgroup 4			
Decap, internal visual design verification	2075	See 3.3.2	
Bond strength	2037	Test condition A; all internal bond wires for each device shall be pulled separately.	
Subgroup 5			
Qualification only			
Thermal impedance	3161	See 4.5.2	
Subgroup 6			
High-temperature (nonoperating) life (LTPD)	1032	T_A = +150°C	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	

1/ For LTPD numbers, see MIL-S-19500.

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TABLE III. Group C inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/	Symbol	Limits Min	Limits Max	Unit
Subgroup 1							
Physical dimensions	2066	See figures 1 and 2					
Subgroup 2							
Thermal shock (glass strain)	1056						
Terminal strength (lead fatigue)	2036	Test condition E					
2N6895 only							
Hermetic seal	1071						
Fine leak Gross leak							
Moisture resistance	1021						
Visual and mechanical evaluation	2071						
Electrical measurement		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					
Subgroup 3							
Shock	2016						
Vibration variable frequency	2056						
Constant acceleration	2006						
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					
Subgroup 4							
Salt atmosphere (corrosion)	1041						
Subgroup 5							
Not applicable							
Subgroup 6							
Intermittent operation life (LTPD)	1042	Test condition D, 6,000 cycles A cycle shall be 30 seconds minimum for the 2N6895 and 1 minute minimum for the 2N6896, 2N6897, and 2N6898.	3				
Hermetic seal	1071						
Fine leak Gross leak							
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8					

See footnote at end of table.

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TABLE III. Group C inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/	Symbol	Limits Min	Limits Max	Unit
Subgroup 7			3				
Qualification only							
Thermal shock temperature cycling destructive	1051	-55°C to 150°C, 200 cycles					
Hermetic seal a. Fine leak b. Gross leak	1071						
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, 7, and 8					
Subgroup 8			10				
Qualification only							
Steady state reverse bias	1042	Test condition A, 1,000 hours.					
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					
Subgroup 9			10				
Qualification only							
Steady state reverse bias	1042	Test condition B, 1,000 hours; V_{GS} = rated voltage					
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7					

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IV. Group A, B, and C electrical end point measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Breakdown voltage drain to source	3407	I_D = -1.0 mA dc; bias condition C; V_{GS} = 0 V	$V_{(BR)DSS}$	-100		V dc
2.	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = -0.25 mA dc	$V_{GS(th)1}$	-2.0	-4.0	V dc
3.	Gate current	3411	V_{GS} = -20 V dc; bias condition C, V_{DS} = 0 V	I_{GSS1}		-100	nA dc
4.	Drain current	3413	V_{DS} = -80 V dc; bias condition C, V_{GS} = 0 V	I_{DSS1}		-25	µA dc
5.	Static drain to source "on" - state resistance"	3421	V_{GS} = -10 V dc; condition A. Pulsed (see 4.5.1)	$r_{DS(on)1}$			
	2N6895		I_D = -0.74 A dc			3.65	ohm
	2N6896		I_D = -3.8 A dc			0.6	ohm
	2N6897		I_D = -7.6 A dc			0.3	ohm
	2N6898		I_D = -15.8 A dc			0.2	ohm
6.	Drain to source "on"-state voltage	3405	V_{GS} = -10 V dc; condition A. Pulsed (see 4.5.1)	$V_{DS(on)}$			
	2N6895		I_D = -1.16 A dc			-6.0	V
	2N6896		I_D = -6.0 A dc			-6.0	V
	2N6897		I_D = -12.0 A dc			-4.8	V
	2N6898		I_D = -25.0 A dc			-6.0	V
7.	Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0	V_{SD}	-0.8	-1.6	V
	2N6895		I_S = -1.16 A dc				
	2N6896		I_S = -6.0 A dc				
	2N6897		I_S = -12.0 A dc				
	2N6898		I_S = -25.0 A dc				
8.	Thermal response	3161	See 4.5.3	ΔV_{SD}	1/		

1/ 10% degradation in group B is permitted.
30% degradation in group C is permitted.

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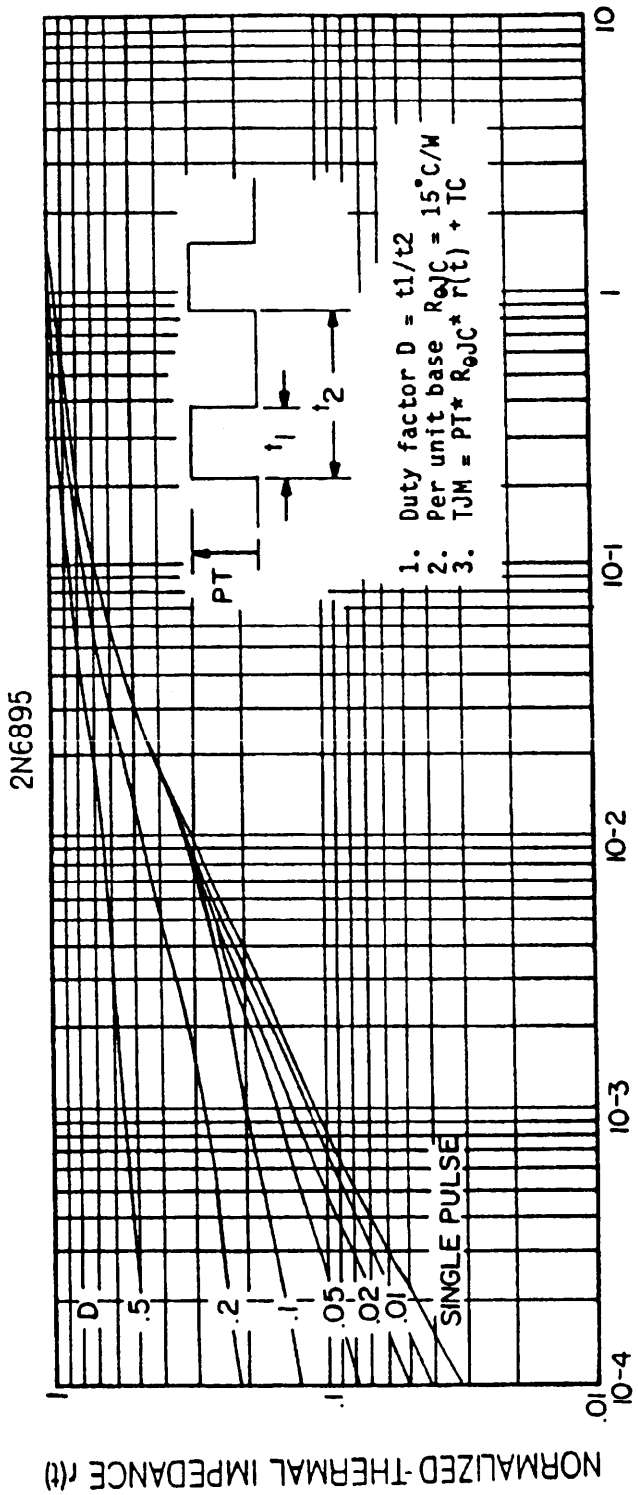


FIGURE 4. Transient thermal response.

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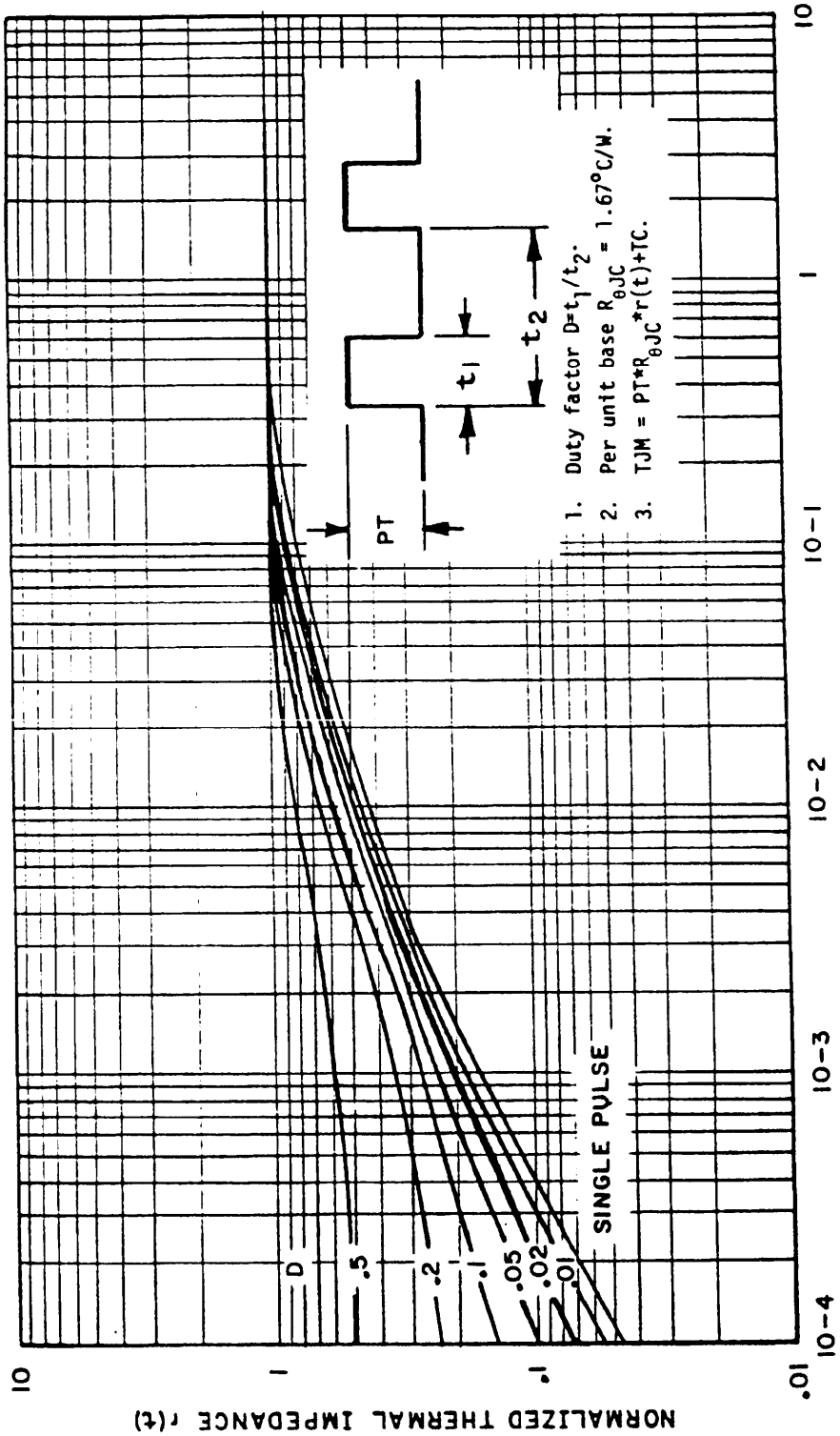


FIGURE 4. Transient thermal response.

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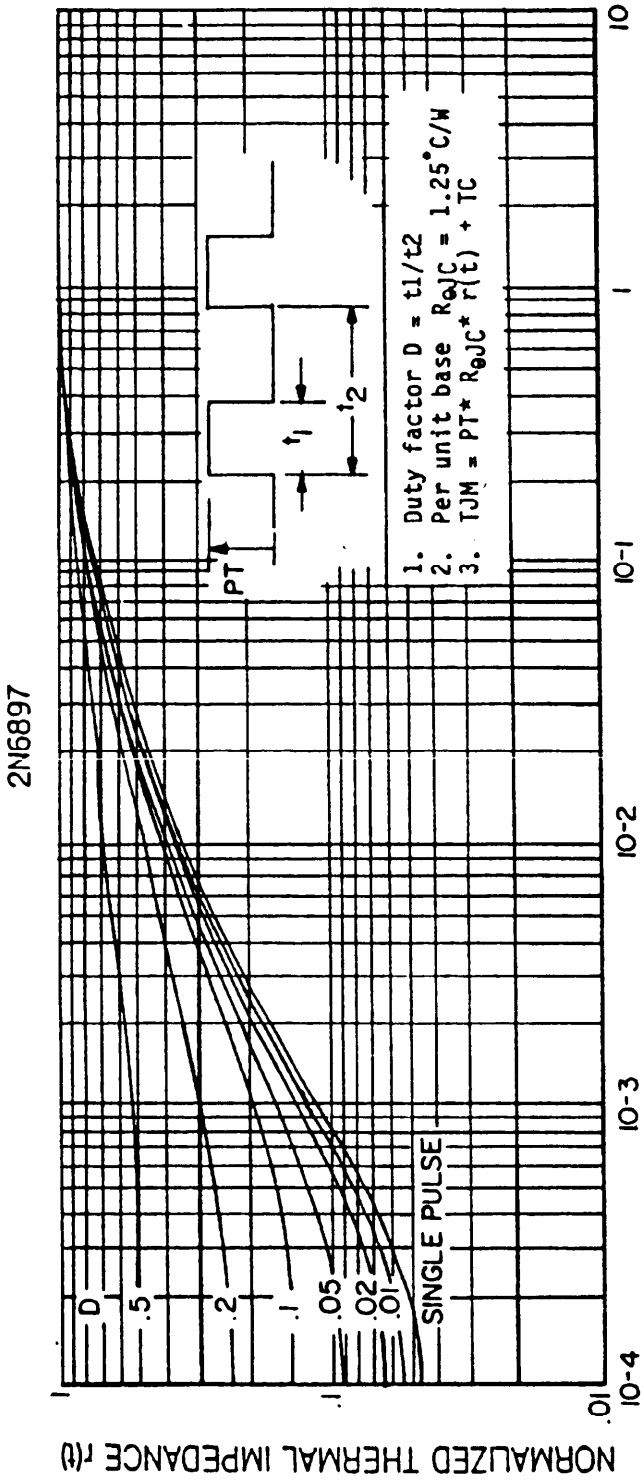


FIGURE 4. Transient thermal response - Continued.

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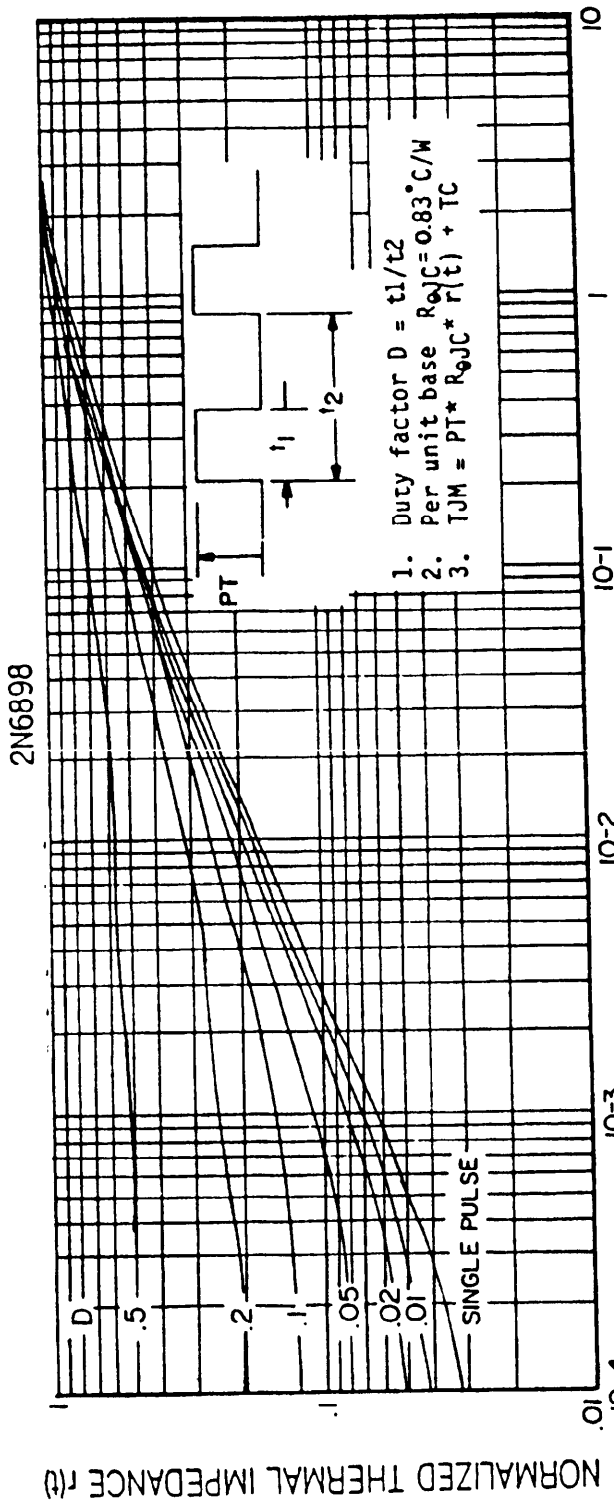


FIGURE 4. Transient thermal response - Continued.

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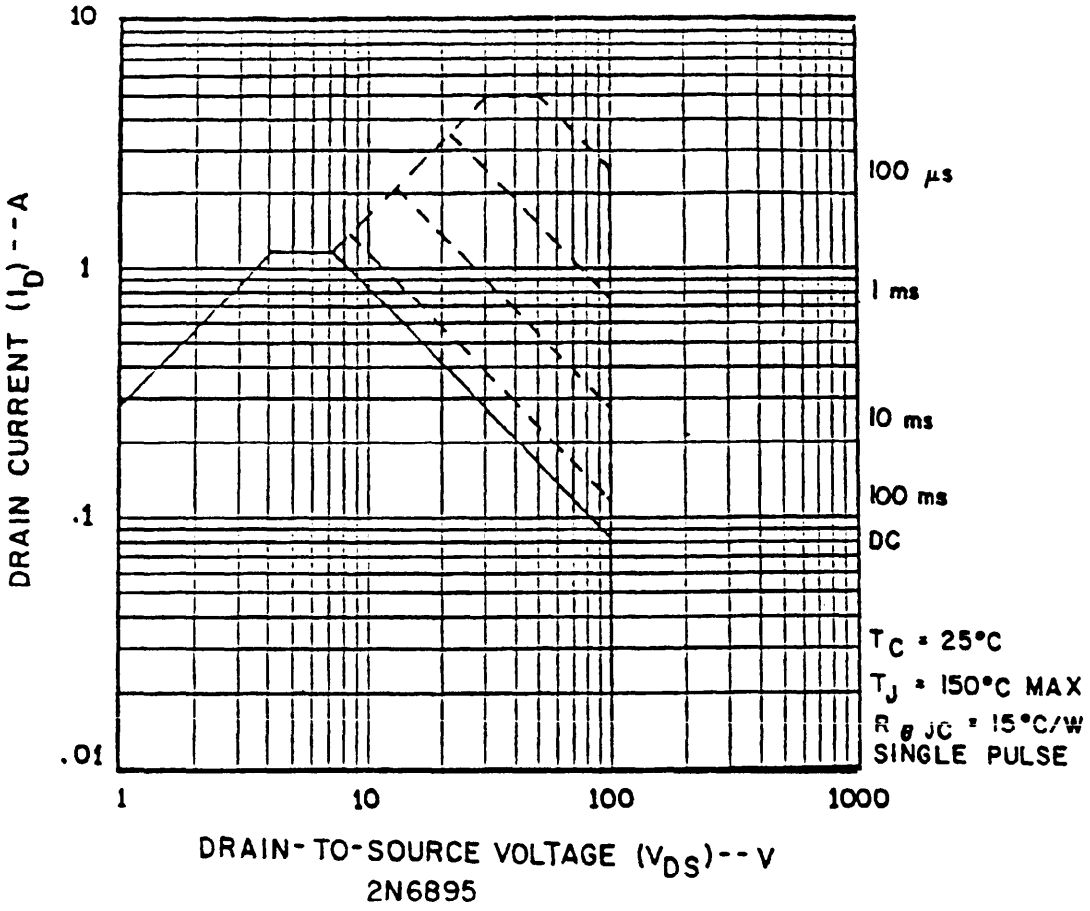
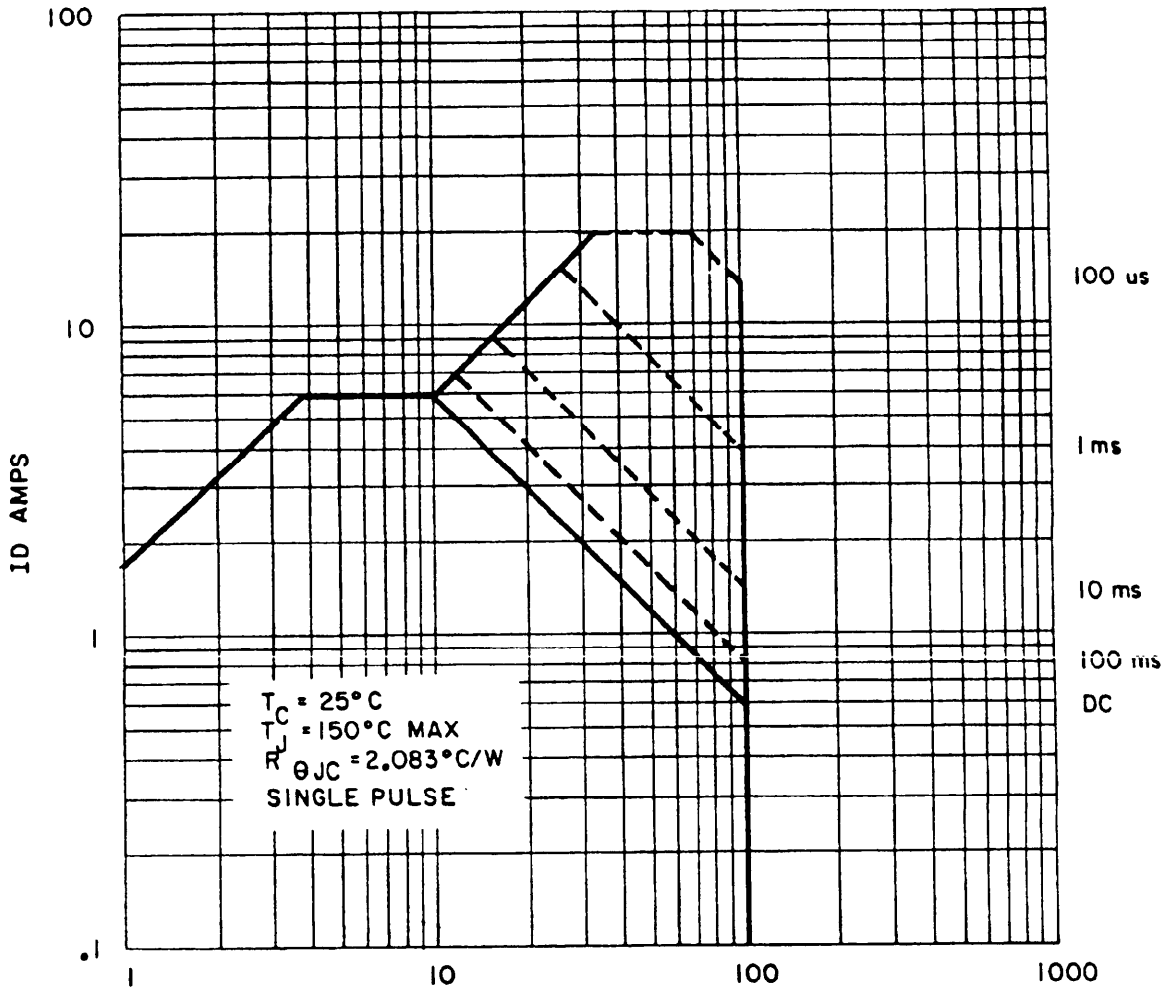


FIGURE 5. Maximum safe operating area.

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MAXIMUM SAFE OPERATING AREA
2N6896

FIGURE 5. Maximum safe operating area.

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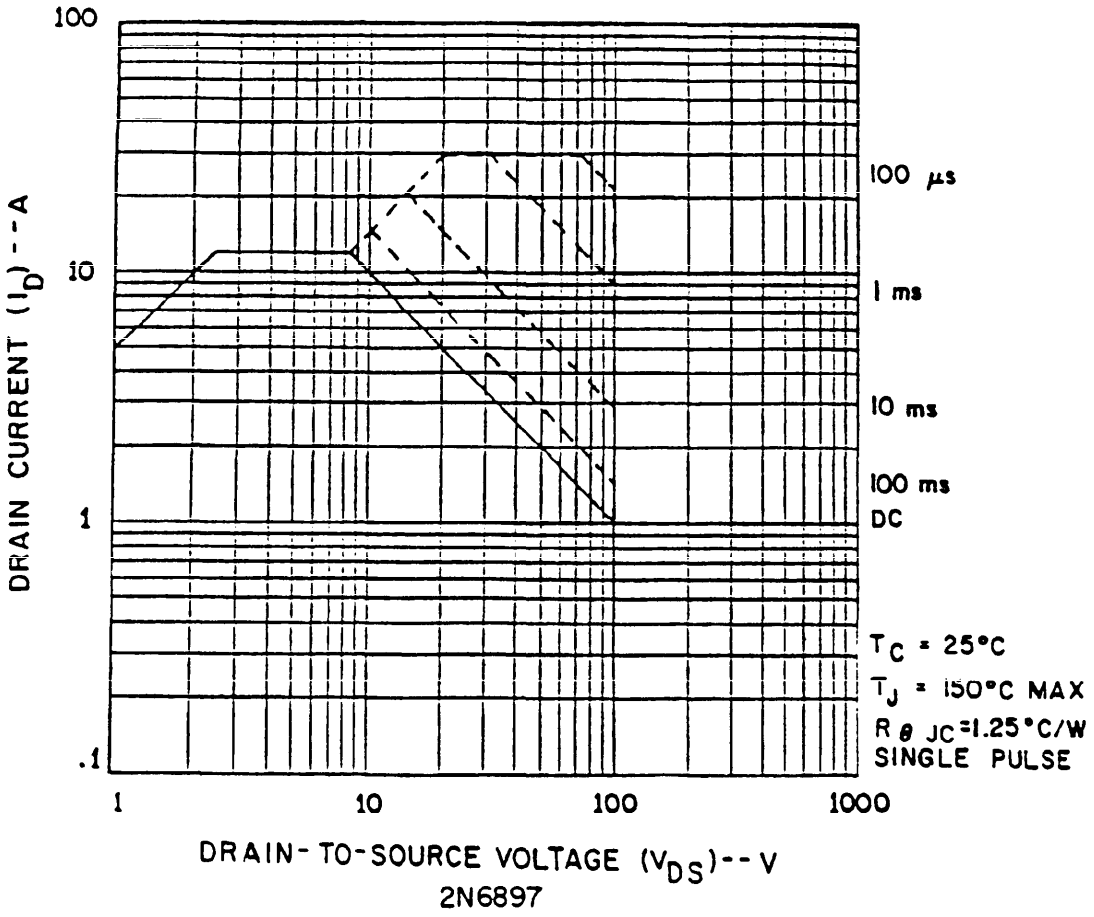
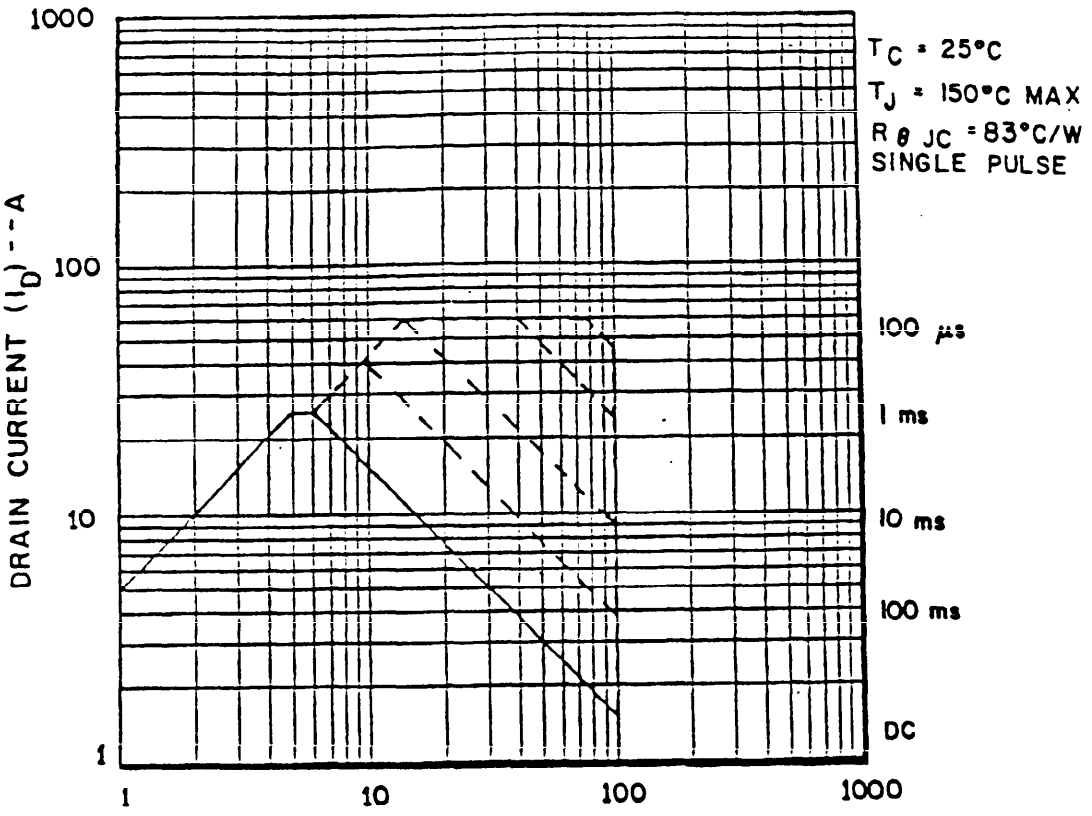


FIGURE 5. Maximum safe operating area - Continued.

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DRAIN-TO-SOURCE VOLTAGE (V_{DS})--V

2N6898

FIGURE 5. Maximum safe operating area - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R $\leq$ 100 K, whenever bias voltage is to be applied drain to source.

6.3 Ordering data. Acquisition documents should specify the lead finish if other than specified (see 3.3.1).

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES
NASA - EB13

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13, 15

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1028)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/565A	Semiconductor Device, Field Effect Transistor, P-Channel, Silicon, Types 2N6895, 2N6896, 2N6897 and 2N6898

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
JANTX, JANTXV and JANS	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

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